Lander Zulueta

Leioa, Bizkaia, Spain · +34 688 725 720 · lander.zulueta@gmail.com
linkedin.com/in/lander-zulueta · github.com/Landerzl · landerzl.github.io

Summary

Physics graduate (four-year degree, ranked 1st in my cohort) and experimental researcher in quantum sensing at Qubiz.team, where I built and operate a fiber-based NV-ensemble magnetometer (CW-ODMR, pulsed Rabi, T_1). Hands-on experience with lasers and optics, cryogenics (LHe/LN_2 handling, sub-Kelvin cryostats), UHV, and microwave/RF design (Ansys HFSS, COMSOL); currently designing a superconducting resonator for CIC nanoGUNE. Experimental work complemented by numerical modeling of NV centers, from open-system dynamics to first-principles DFT of their response to electric fields.

Education

B.Sc. in Physics (4 years, 240 ECTS) 2022 – 2026
University of the Basque Country (EHU) *Leioa, Spain*

- Grade: **9.06/10**, **ranked 1st** in the Physics degree.
- Coursework: Quantum Mechanics, Thermodynamics, Statistical Mechanics, Electromagnetism, Solid State Physics, Advanced Laboratory Techniques.
- **Thesis** (10/10, Honors): *Critical Metrology* (see Research Experience).

Research Experience

Full-Time Researcher, Quantum Sensing Sep. 2026 – present
Qubiz.team (quantum technology startup co-founded by Prof. Jorge Casanova, EHU) *Bilbao, Spain*

- Built and operate a fiber-based NV-ensemble setup (CVD diamond) for magnetic field sensing: 520 nm laser excitation with TTL modulation, APD detection and lock-in readout.
- Measured CW-ODMR (∼1.5% contrast), pulsed Rabi oscillations and T_1 relaxation.
- Designed and simulated microwave waveguides for hyperpolarization setups with Ansys HFSS and COMSOL RF.

Superconducting Resonator Design, Quantum Probe Microscopy Group Oct. 2026 – present
CIC nanoGUNE (Prof. Nacho Pascual) *San Sebastián, Spain*

- Designing and simulating a 4.5 GHz superconducting resonator (proximitized Au/Nb) that will serve as the STM sample, with the tip tunneling directly onto the resonator.

Research Intern, Quantum Probe Microscopy Group Jun. – Aug. 2026
CIC nanoGUNE (Supervisor: Prof. Nacho Pascual) *San Sebastián, Spain*

- Operated a low-temperature STM (LT-STM), including LHe/LN_2 transfers and sub-Kelvin cryostats.
- Characterized the transmission function of a cryogenic microwave antenna integrated into the STM junction, laying the RF calibration groundwork for ESR-STM measurements.
- Prepared STM tips and Au(111) substrates (UHV annealing, sputtering, MBE) and synthesized graphene nanoribbons by on-surface synthesis, verified by STM topography.

Researcher, Quantum Sensing Jun. – Dec. 2025
Qubiz.team *Bilbao, Spain*

- Contributed to early-stage projects in biomedicine, automotive and electrical industries, studying quantum sensing devices with a focus on NV centers.
- Built RF components of a bench-top NMR device and tested classical machine-learning methods to enhance its signal.

Undergraduate Researcher (Bachelor's Thesis) Sep. 2025 – Jul. 2026
EHU (Supervisors: Iñigo Egusquiza, Jagoba Apellaniz; Ikasiker grant) *Leioa, Spain*

- *Critical Metrology.* Question: can quantum metrology gain from operating near a quantum phase transition? Method: used the superradiant transition of the quantum Rabi model in the classical-oscillator limit. Result: showed that the precision of parameter estimation diverges at the critical point in the static limit.
- Analytic work with Schrieffer-Wolff and bosonic Bogoliubov transformations; numerics with dense diagonalization and numerical differentiation.
- Code: github.com/Landerzl/Critical-Metrology

Selected Projects

- **Electric fields on the NV center** (Sep. 2026 – present). Own numerical study, started at the suggestion of Prof. Jorge Casanova, of how a DC electric field couples to the NV^- spin.
  - Derived the field–spin couplings allowed by C_{3v} symmetry and time reversal in the ground ($^3A_2$) and excited (^3E) states, to first and second order in the field.
  - Finite-field unrestricted DFT (PySCF, PBE/B3LYP) on hydrogen-terminated diamond clusters up to $C_{163}NH_{100}^-$, with the field included self-consistently. Excited state with ΔSCF/IMOM (vertical excitation 1.88 eV).
  - Fed the couplings into a Lindblad model: predicted an ESLAC shift of ∼3.3 G per MV/cm for an axial field, and proposed a lock-in measurement using field-polarity modulation.
- **NV open-system dynamics.** Seven-level model with the Lindblad equation (QuTiP) reproducing CW-ODMR, Ramsey interferometry and the photoluminescence around the GSLAC and ESLAC as a function of magnetic and electric field. Part of the code: github.com/Landerzl/NV-Centers-Code.
- **BB84 QKD experiment.** Built an experiment at EHU to exchange messages with the BB84 protocol, including the free-space optical alignment and operation of the hardware; demonstrated it live to undergraduate students.

Honors and Awards

- **Extraordinary Degree Award in Physics**, EHU (2026): best academic record of the 2025–26 cohort.
- **Ikasiker grant**, Basque Government (2026): competitive grant for the development of the Bachelor's Thesis.
- **Honors in 15 subjects** (2022–2026): Bachelor's Thesis (Critical Metrology), Solid State Physics I & II, Quantum Physics, Nuclear and Particle Physics, Experimental Techniques II, III & IV, Sensors and Actuators, Electromagnetism II, Modern Physics, Vector and Complex Analysis, Algebra I, Chemistry II, Fundamentals of Programming.
- **Academic Acceleration** (2021): skipped one year of secondary school for outstanding academic performance.

Courses and Certifications

- Laser Safety Officer (16h), ProCareLight, Sep. 2026
- Quantum Machine Learning and Quantum Optimization (36h), University of Deusto, Nov. 2026 (upcoming)
- Quantum Chemistry with VQE (5h), IBM, Sep. 2026
- Introduction to Ansys HFSS (2h), Ansys, Sep. 2026
- Quantum Communication and Quantum Networks (4h), TU Braunschweig, Feb. 2026
- General Formulation of Quantum Information (10h), IBM, Nov. 2025
- Fundamentals of Quantum Algorithms (10h), IBM, Aug. 2025
- Quantum Sensing and Imaging (30h), Qureca, Jul. 2025

Seminars and Workshops

- Hands on Quantum Information (3 ECTS), EHU Quantum Center, Mar. 2026 – present
- 8th Basque Quantum Science and Technology Workshop / 4th BasQ/IKUR Quantum Science and Technology Workshop, Oct. 2025
- Quantum Computing: from Fundamentals to Applications (1 ECTS), BasQ, DIPC, Sep. 2025
- Quantum Technologies: Present and Future, Nquire, Jun. 2025

Technical Skills

Experimental: Lasers and optics (free-space alignment, fiber coupling, TTL laser modulation, laser safety), cryogenics (LHe/LN_2, sub-Kelvin cryostats), UHV, microwave/RF delivery, NV-ensemble sensing (CW-ODMR, Rabi, T_1), low-temperature STM
Programming: Python (QuTiP, scqubits, PySCF, NumPy, SciPy), MATLAB, Wolfram Mathematica, Fortran 90
Simulation and RF: COMSOL Multiphysics, Ansys HFSS, Qorvo MatchCalc, Smith charts, LAMMPS, Fusion 360, PSpice
Tools: LaTeX, Git/GitHub, LabVIEW, Linux (WSL)
Languages: Basque (native), Spanish (native), English (C1, Cambridge, Sep. 2025), French (basic)

References

Prof. Nacho Pascual, CIC nanoGUNE (internship supervisor)
Prof. Iñigo Egusquiza, University of the Basque Country (EHU) (thesis supervisor)
Prof. Jorge Casanova, University of the Basque Country (EHU) and co-founder of Qubiz.team (employer)